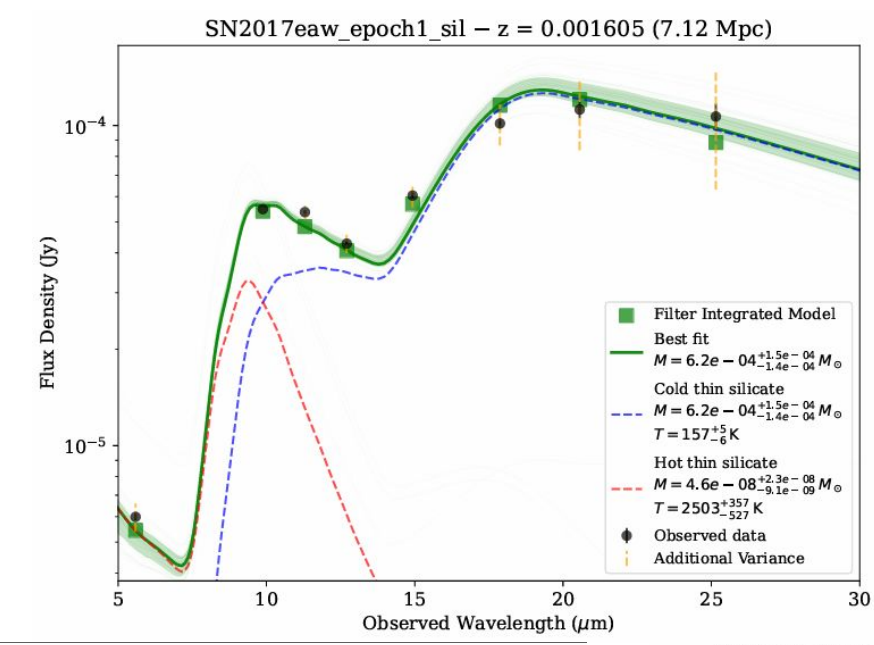
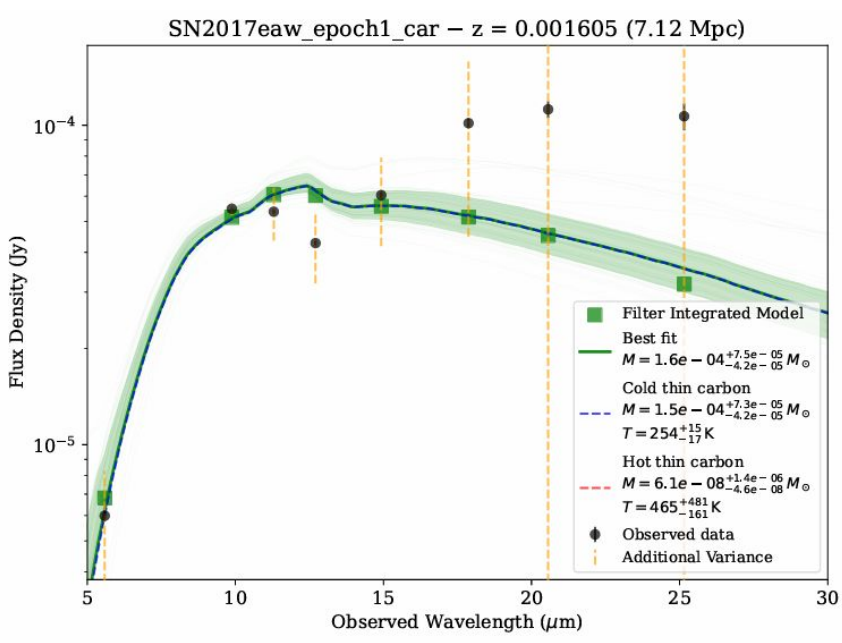
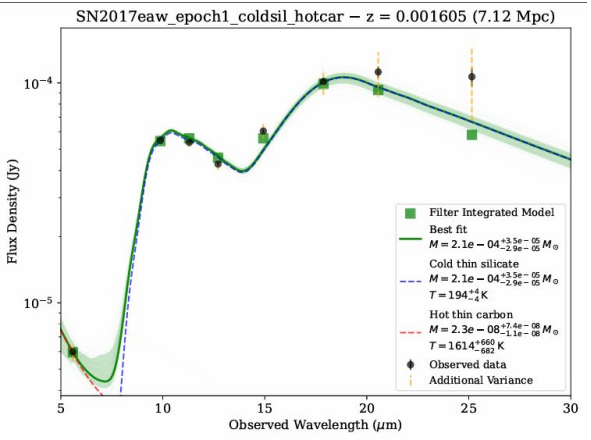
DustySN Graphs

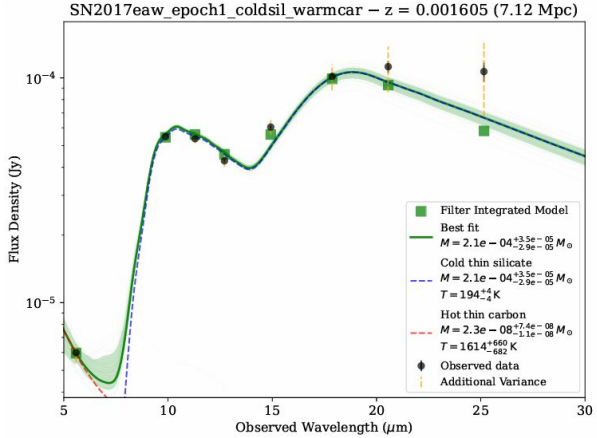


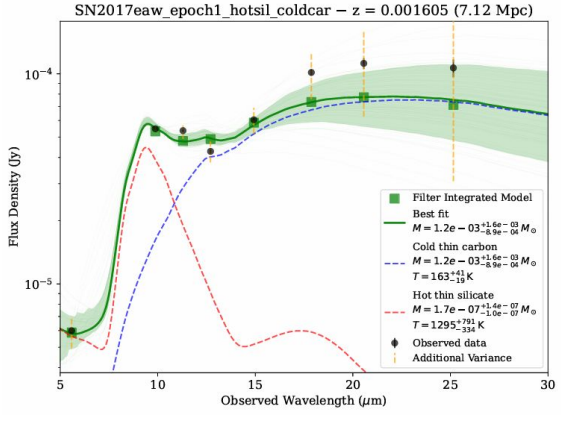

2017eaw epoch 1


SN2017eaw_epoch1_sil − z = 0.001605 (7.12 Mpc)
Flux Density (Jy)
Observed Wavelength (μm)
Filter Integrated Model
Best fit
$M = 6.2e-04^{+1.5e-04}_{-1.4e-04}\,M_\odot$
Cold thin silicate
$M = 6.2e-04^{+1.5e-04}_{-1.4e-04}\,M_\odot$
$T = 157^{+5}_{-6}$ K
Hot thin silicate
$M = 4.6e-08^{+2.3e-08}_{-9.1e-09}\,M_\odot$
$T = 2503^{+357}_{-527}$ K
Observed data
Additional Variance

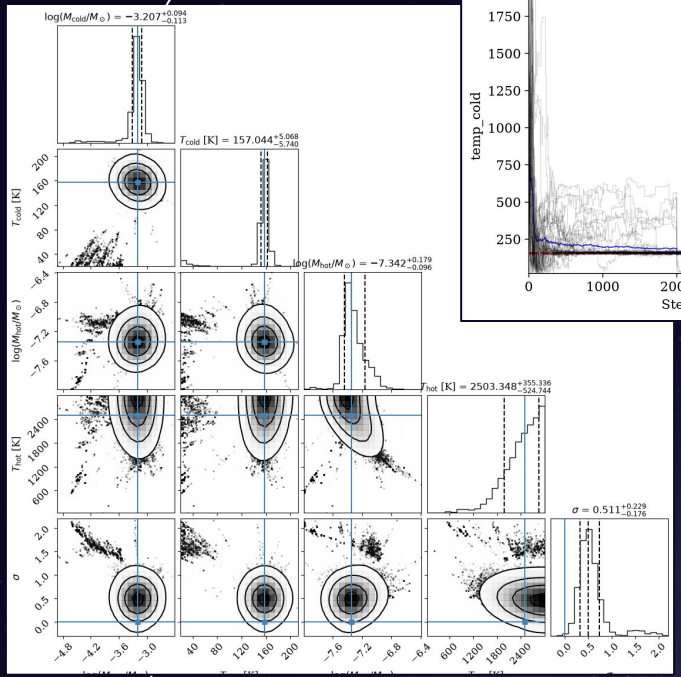

SN2017eaw_epoch1_car − z = 0.001605 (7.12 Mpc)
Flux Density (Jy)
Observed Wavelength (μm)
Filter Integrated Model
Best fit
$M = 1.6e-04^{+7.5e-05}_{-4.2e-05}\,M_\odot$
Cold thin carbon
$M = 1.5e-04^{+7.3e-05}_{-4.2e-05}\,M_\odot$
$T = 254^{+15}_{-17}$ K
Hot thin carbon
$M = 6.1e-08^{+1.4e-06}_{-4.6e-08}\,M_\odot$
$T = 465^{+481}_{-161}$ K
Observed data
Additional Variance

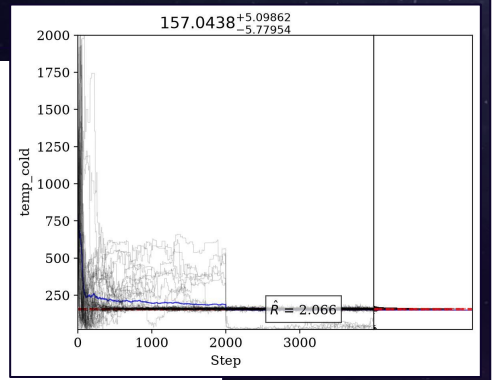

SN2017eaw_epoch1_coldsil_hotcar − z = 0.001605 (7.12 Mpc)
Flux Density (Jy)
Observed Wavelength (μm)
Filter Integrated Model
Best fit
$M = 2.1e-04^{+3.5e-05}_{-2.9e-05}\,M_\odot$
Cold thin silicate
$M = 2.1e-04^{+3.5e-05}_{-2.9e-05}\,M_\odot$
$T = 194^{+4}_{-4}$ K
Hot thin carbon
$M = 2.3e-08^{+7.4e-08}_{-1.1e-08}\,M_\odot$
$T = 1614^{+660}_{-682}$ K
Observed data
Additional Variance

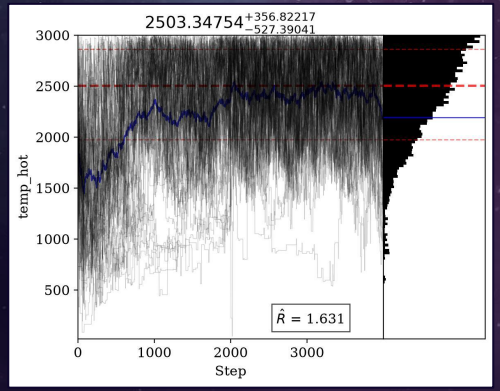

SN2017eaw_epoch1_coldsil_warmcar − z = 0.001605 (7.12 Mpc)
Flux Density (Jy)
Observed Wavelength (μm)
Filter Integrated Model
Best fit
$M = 2.1e-04^{+3.5e-05}_{-2.9e-05}\,M_\odot$
Cold thin silicate
$M = 2.1e-04^{+3.5e-05}_{-2.9e-05}\,M_\odot$
$T = 194^{+4}_{-4}$ K
Hot thin carbon
$M = 2.3e-08^{+7.4e-08}_{-1.1e-08}\,M_\odot$
$T = 1614^{+660}_{-682}$ K
Observed data
Additional Variance

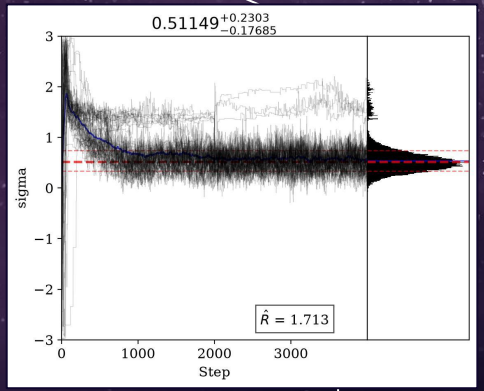

SN2017eaw_epoch1_hotsil_coldcar − z = 0.001605 (7.12 Mpc)
Flux Density (Jy)
Observed Wavelength (μm)
Filter Integrated Model
Best fit
$M = 1.2e-03^{+1.6e-03}_{-8.9e-04}\,M_\odot$
Cold thin carbon
$M = 1.2e-03^{+1.6e-03}_{-8.9e-04}\,M_\odot$
$T = 163^{+41}_{-19}$ K
Hot thin silicate
$M = 1.7e-07^{+1.4e-07}_{-1.0e-07}\,M_\odot$
$T = 1295^{+791}_{-334}$ K
Observed data
Additional Variance

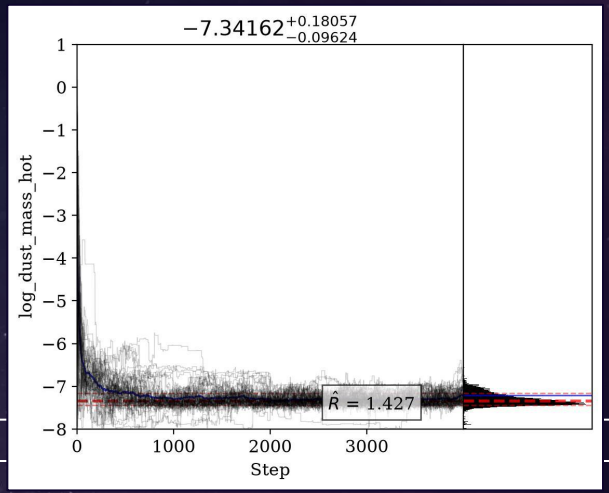
2-Component Silicate

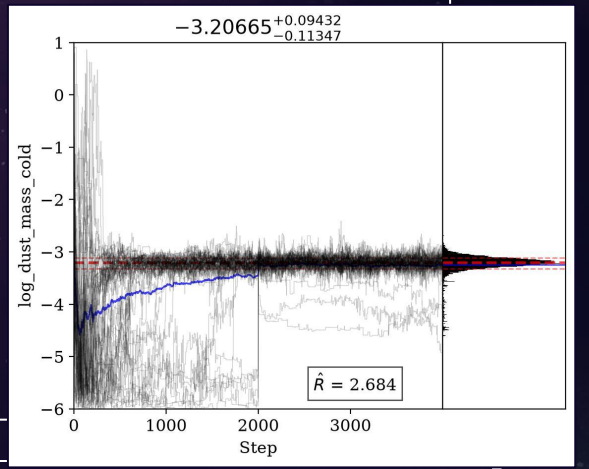




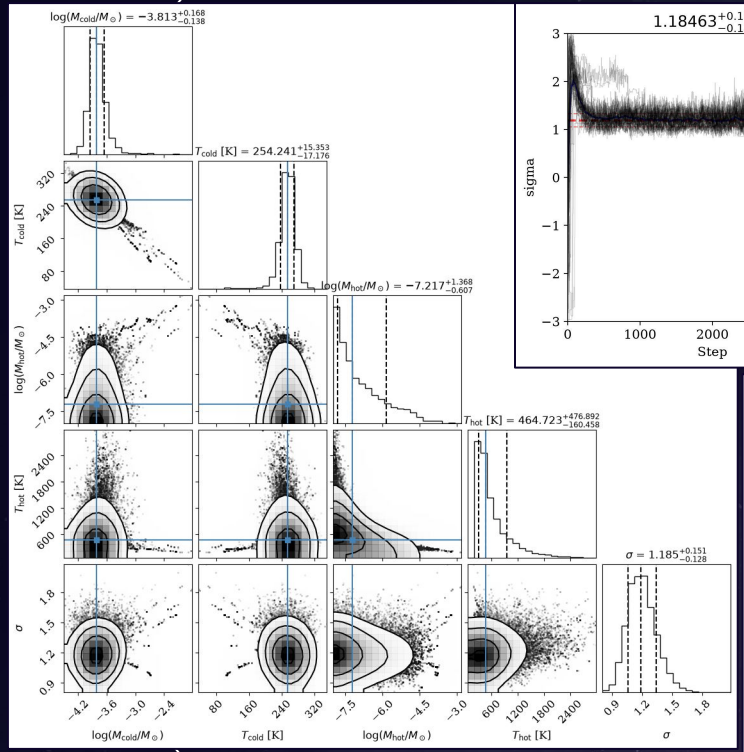


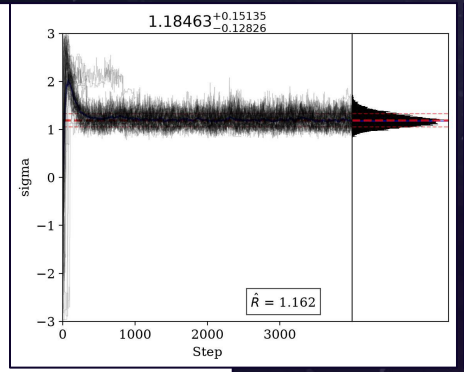


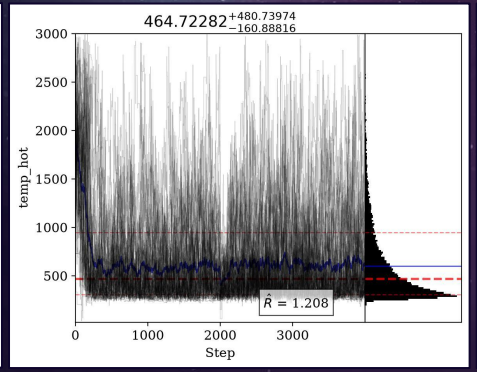


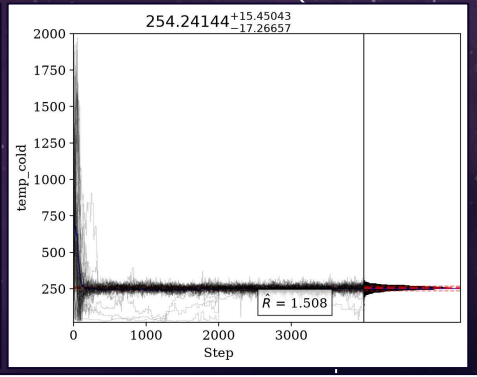


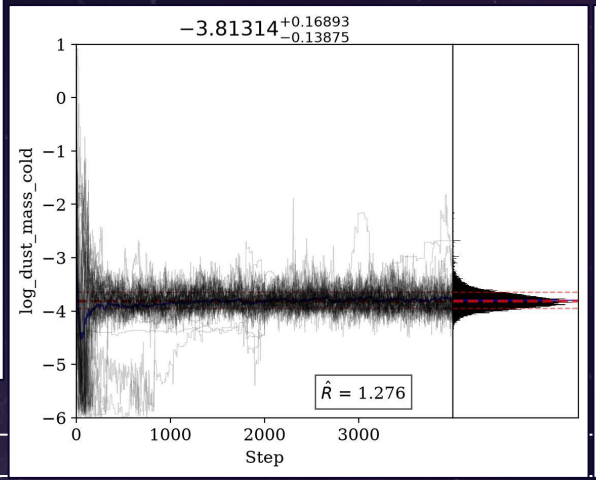
2-Component Carbon/Graphite

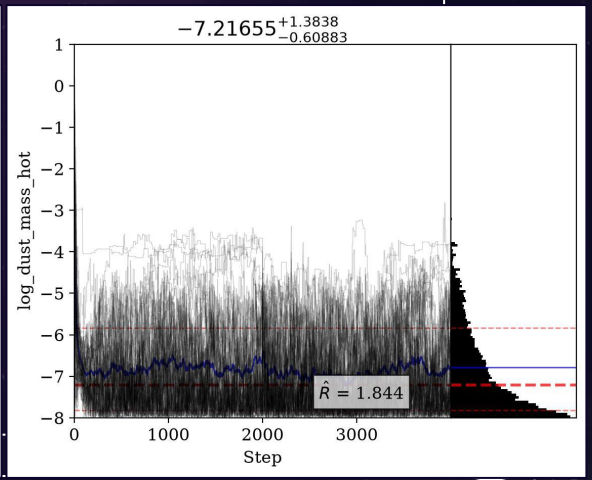




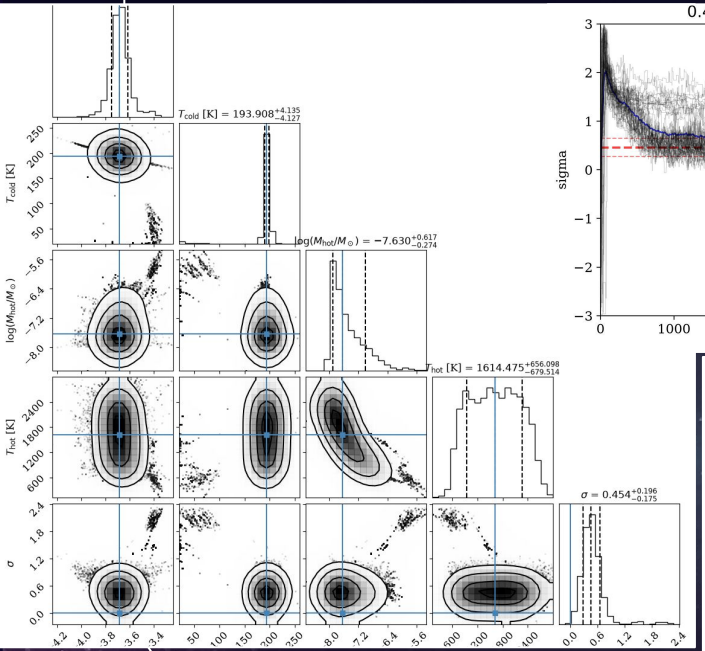


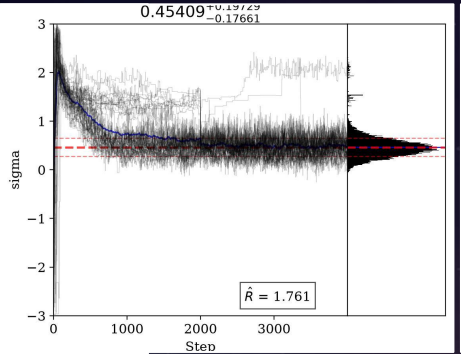


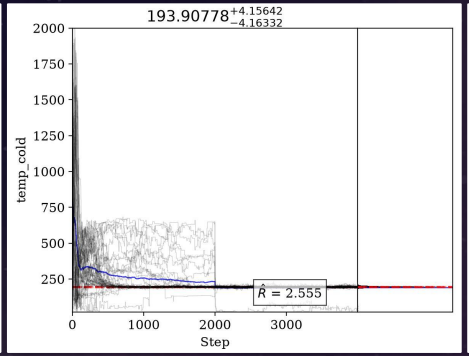


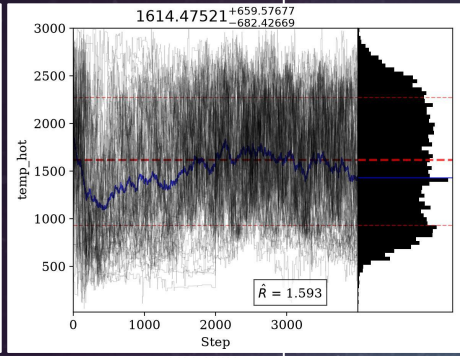


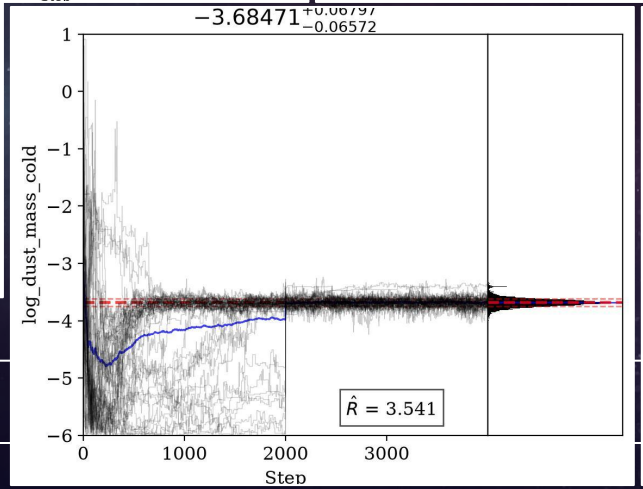
Cold Silicate, Hot Carbon

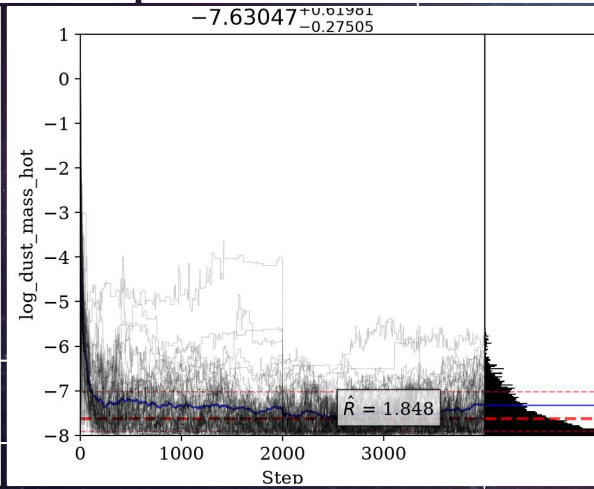




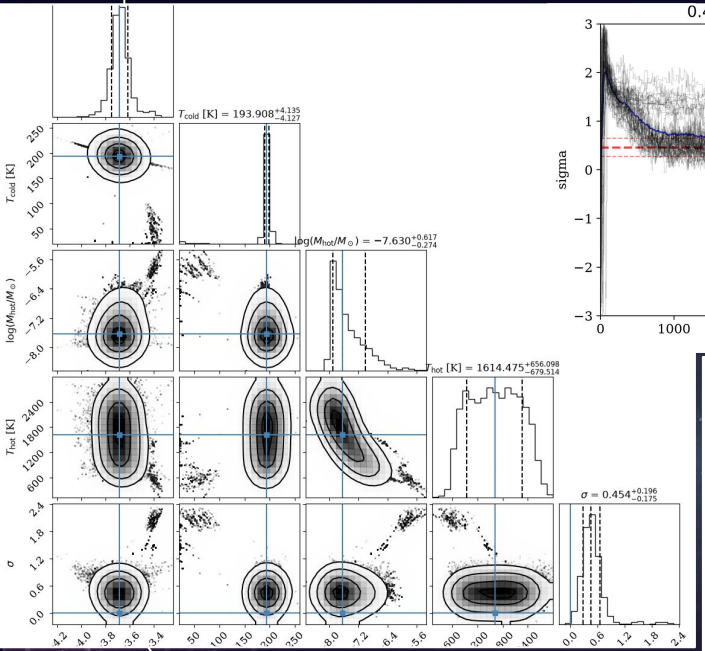


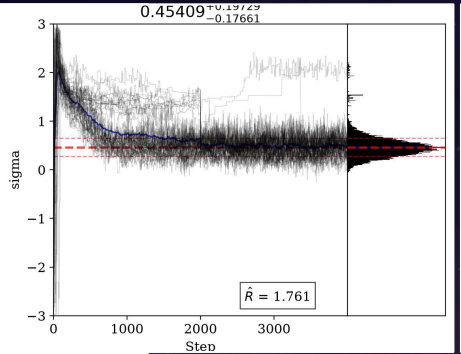


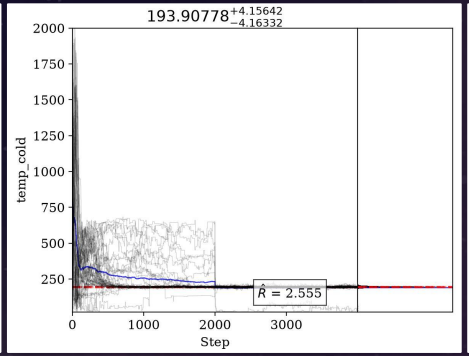


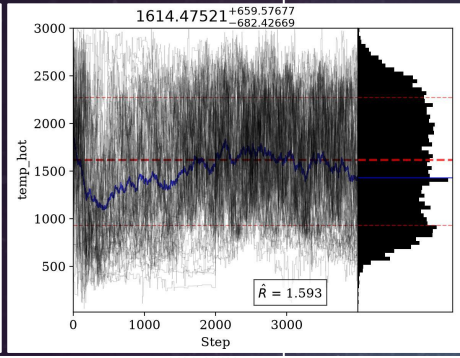


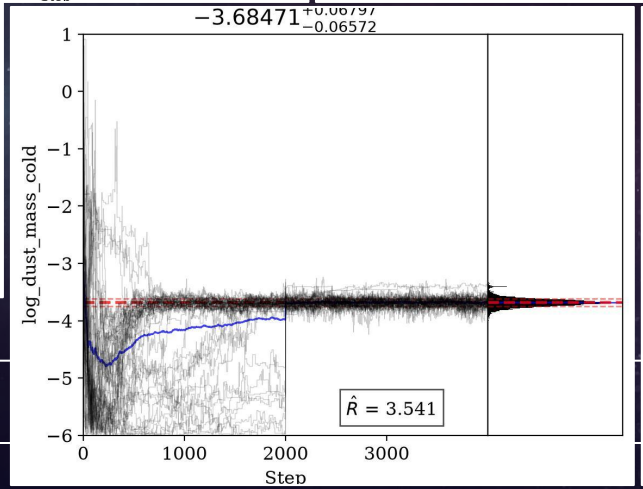
Cold Silicate, Warm Carbon

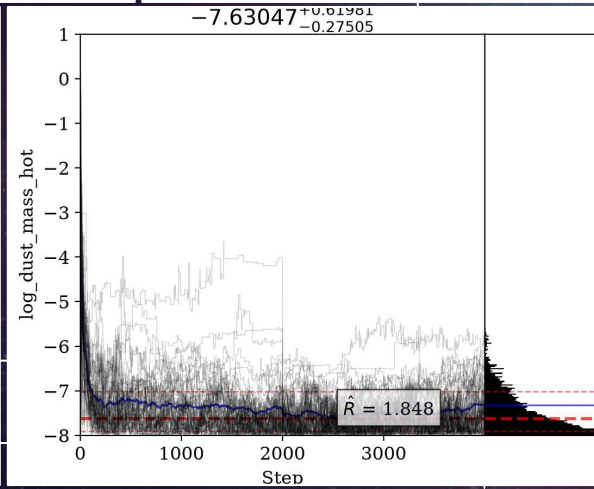




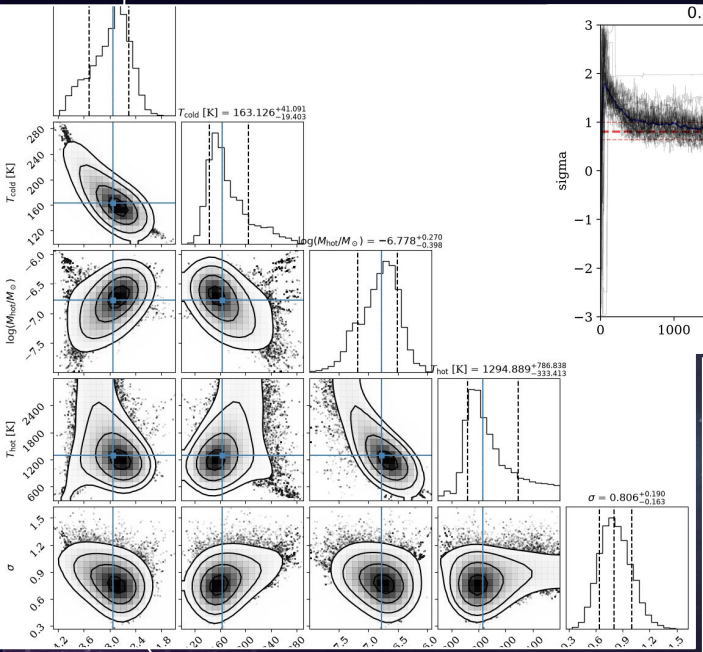


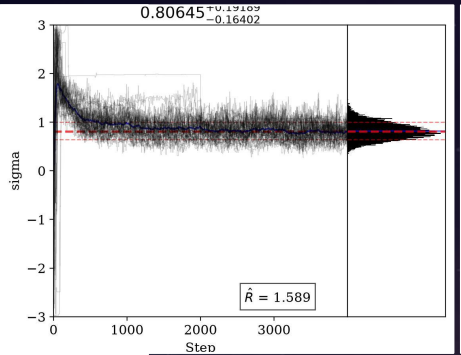


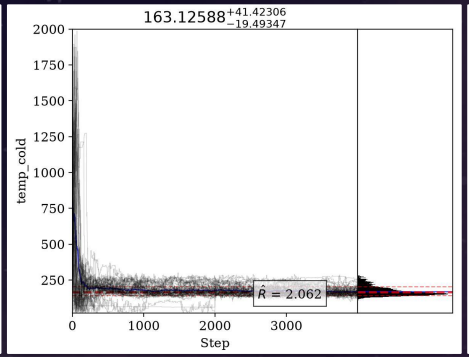


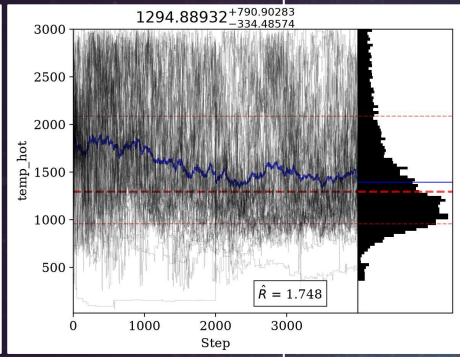


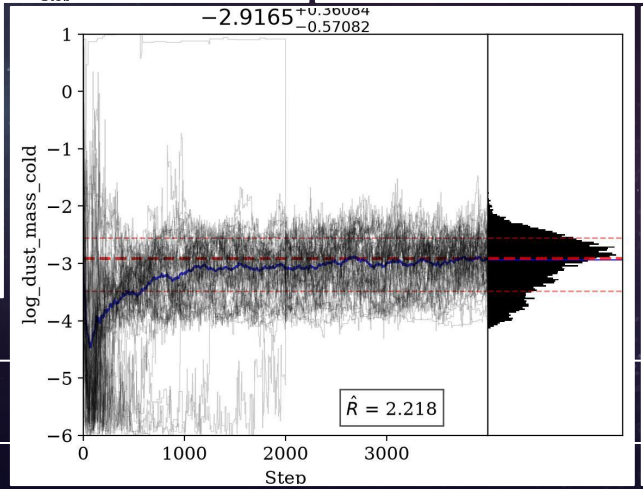
Hot Silicate, Cold Carbon

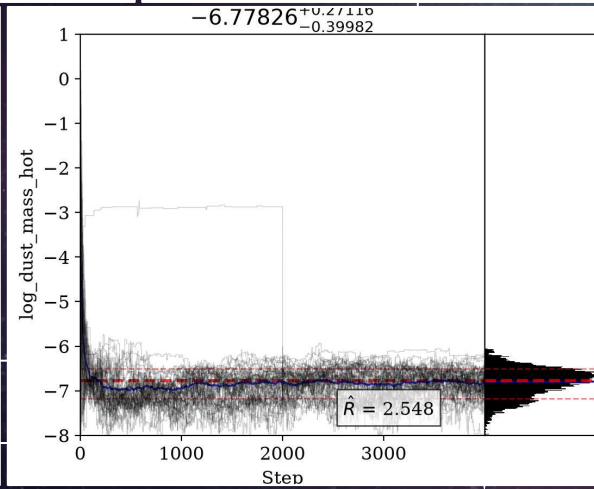




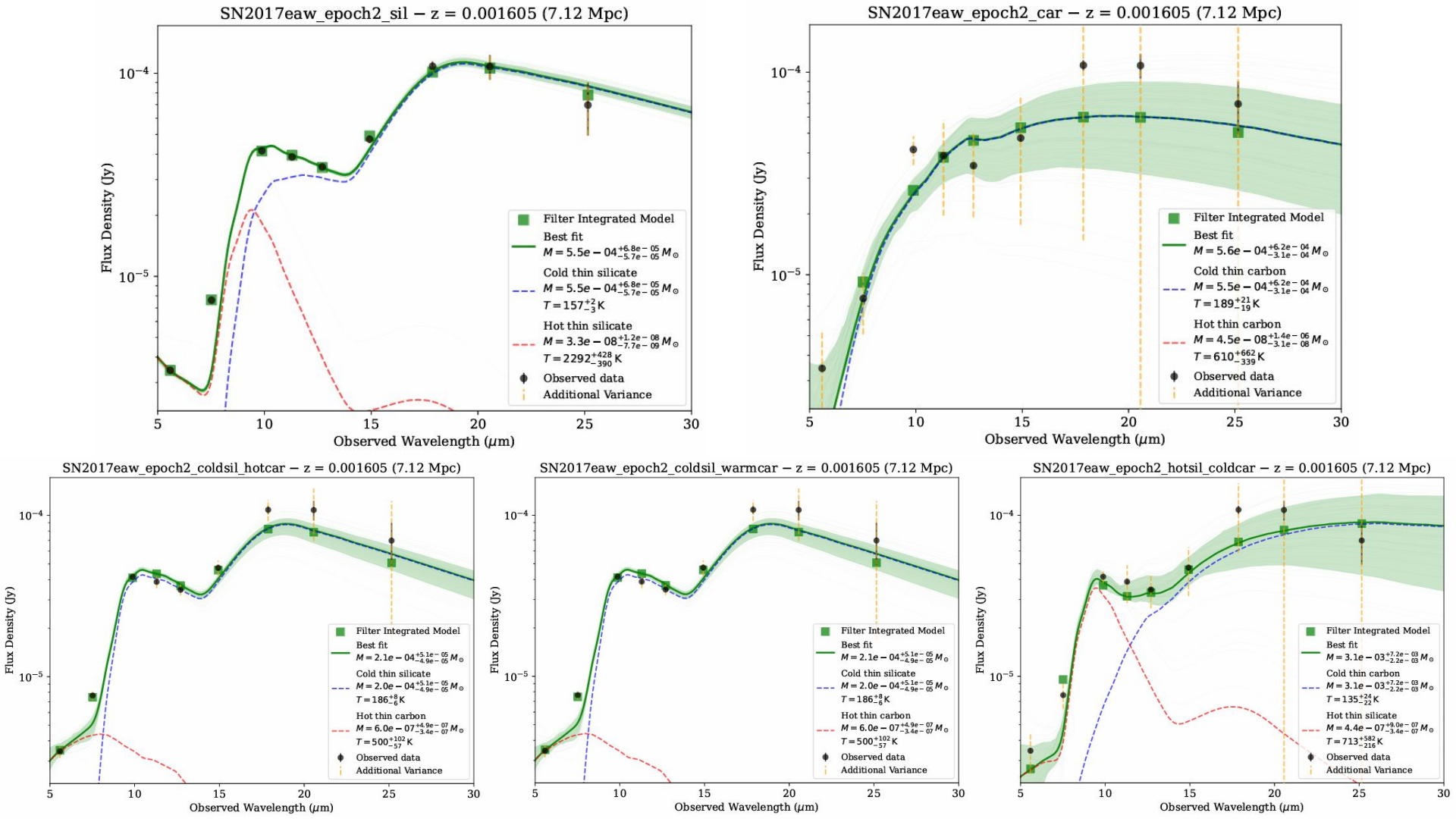




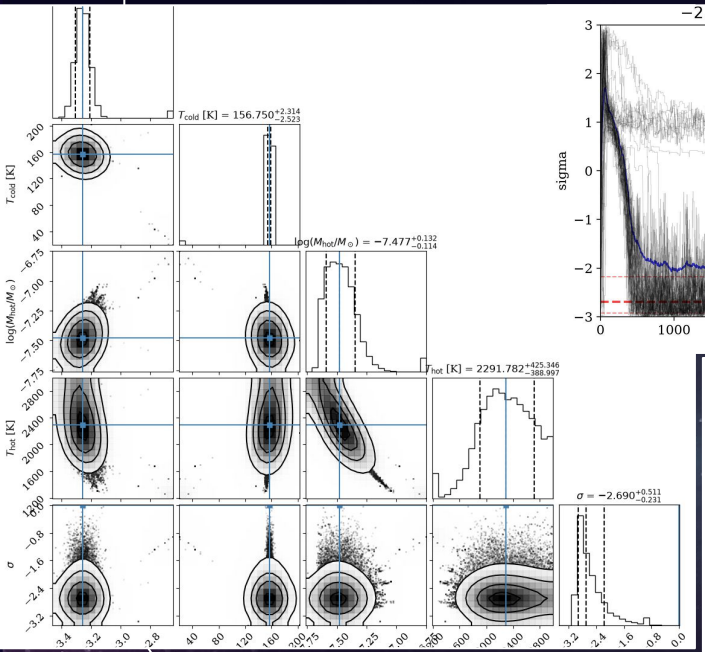


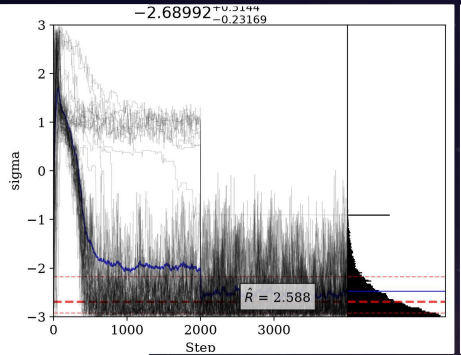


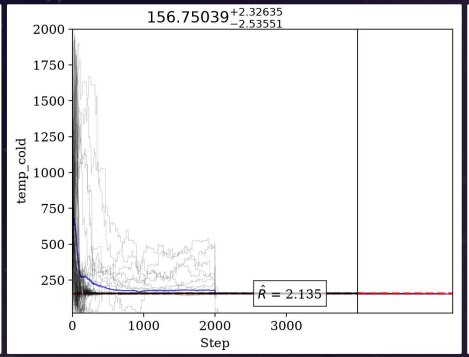

2017eaw epoch 2

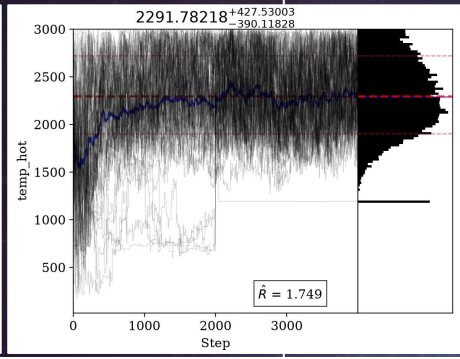

SN2017eaw_epoch2_sil − z = 0.001605 (7.12 Mpc)
Flux Density (Jy)
Observed Wavelength (μm)
Filter Integrated Model
Best fit
$M = 5.5e-04^{+6.8e-05}_{-5.7e-05}\,M_\odot$
Cold thin silicate
$M = 5.5e-04^{+6.8e-05}_{-5.7e-05}\,M_\odot$
$T = 157^{+2}_{-3}$ K
Hot thin silicate
$M = 3.3e-08^{+1.2e-08}_{-7.7e-09}\,M_\odot$
$T = 2292^{+428}_{-390}$ K
Observed data
Additional Variance
SN2017eaw_epoch2_car − z = 0.001605 (7.12 Mpc)
Flux Density (Jy)
Observed Wavelength (μm)
Filter Integrated Model
Best fit
$M = 5.6e-04^{+6.2e-04}_{-3.1e-04}\,M_\odot$
Cold thin carbon
$M = 5.5e-04^{+6.2e-04}_{-3.1e-04}\,M_\odot$
$T = 189^{+21}_{-19}$ K
Hot thin carbon
$M = 4.5e-08^{+1.4e-06}_{-3.1e-08}\,M_\odot$
$T = 610^{+662}_{-339}$ K
Observed data
Additional Variance
SN2017eaw_epoch2_coldsil_hotcar − z = 0.001605 (7.12 Mpc)
Flux Density (Jy)
Observed Wavelength (μm)
Filter Integrated Model
Best fit
$M = 2.1e-04^{+5.1e-05}_{-4.9e-05}\,M_\odot$
Cold thin silicate
$M = 2.0e-04^{+5.1e-05}_{-4.9e-05}\,M_\odot$
$T = 186^{+8}_{-6}$ K
Hot thin carbon
$M = 6.0e-07^{+4.9e-07}_{-3.4e-07}\,M_\odot$
$T = 500^{+102}_{-57}$ K
Observed data
Additional Variance
SN2017eaw_epoch2_coldsil_warmcar − z = 0.001605 (7.12 Mpc)
Flux Density (Jy)
Observed Wavelength (μm)
Filter Integrated Model
Best fit
$M = 2.1e-04^{+5.1e-05}_{-4.9e-05}\,M_\odot$
Cold thin silicate
$M = 2.0e-04^{+5.1e-05}_{-4.9e-05}\,M_\odot$
$T = 186^{+8}_{-6}$ K
Hot thin carbon
$M = 6.0e-07^{+4.9e-07}_{-3.4e-07}\,M_\odot$
$T = 500^{+102}_{-57}$ K
Observed data
Additional Variance
SN2017eaw_epoch2_hotsil_coldcar − z = 0.001605 (7.12 Mpc)
Flux Density (Jy)
Observed Wavelength (μm)
Filter Integrated Model
Best fit
$M = 3.1e-03^{+7.2e-03}_{-2.2e-03}\,M_\odot$
Cold thin carbon
$M = 3.1e-03^{+7.2e-03}_{-2.2e-03}\,M_\odot$
$T = 135^{+24}_{-22}$ K
Hot thin silicate
$M = 4.4e-07^{+9.0e-07}_{-3.4e-07}\,M_\odot$
$T = 713^{+582}_{-216}$ K
Observed data
Additional Variance

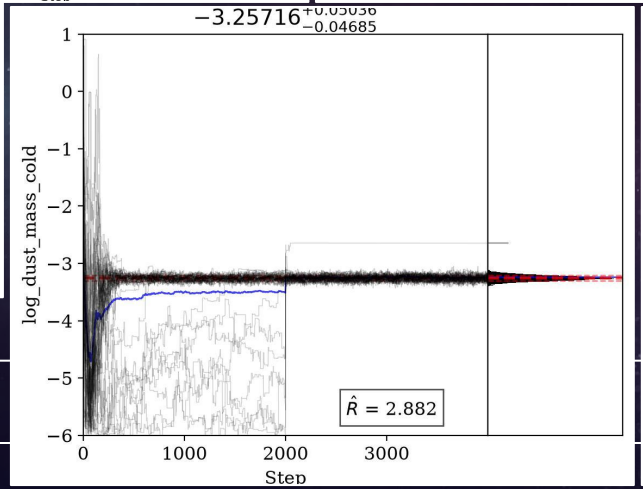
2C Silicate

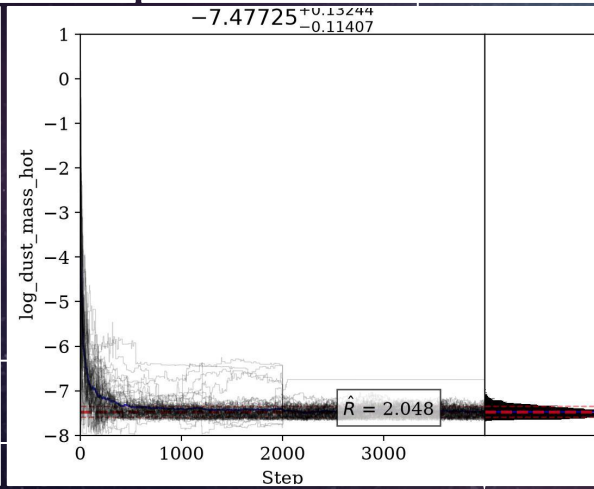

T_{cold} [K] = $156.750^{+2.314}_{-2.523}$
$\log(M_{hot}/M_\odot) = -7.477^{+0.132}_{-0.114}$
T_{hot} [K] = $2291.782^{+425.346}_{-388.997}$
$\sigma = -2.690^{+0.511}_{-0.231}$
T_{cold} [K]
$\log(M_{hot}/M_\odot)$
T_{hot} [K]
σ


$-2.68992^{+0.3144}_{-0.23169}$
sigma
$\hat{R}$ = 2.588
Step

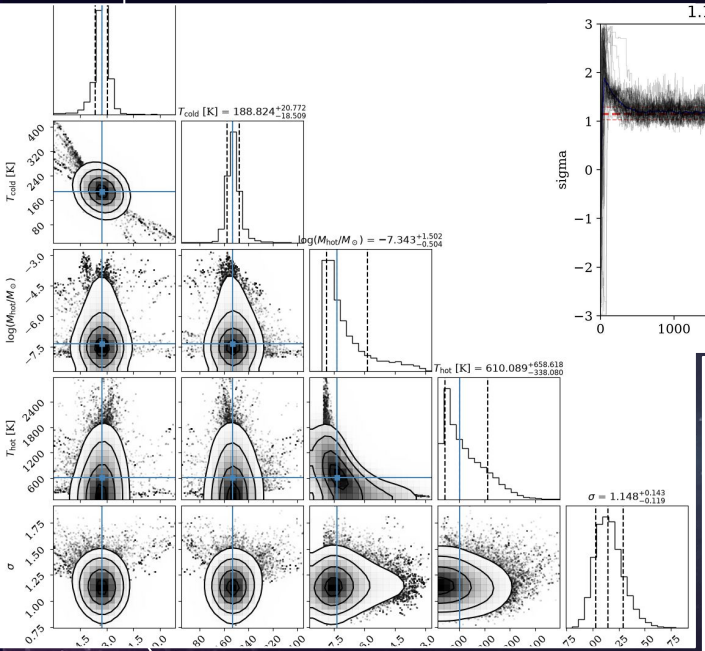

$156.75039^{+2.32635}_{-2.53551}$
temp_cold
$\hat{R}$ = 2.135
Step

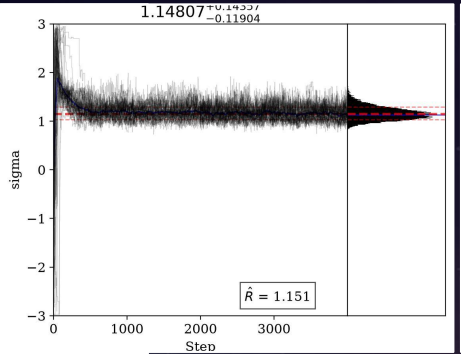

$2291.78218^{+427.53003}_{-390.11828}$
temp_hot
$\hat{R}$ = 1.749
Step

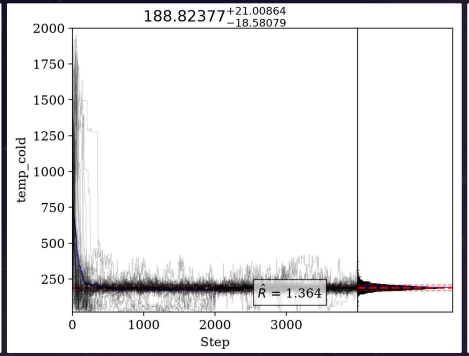

$-3.25716^{+0.05036}_{-0.04685}$
log_dust_mass_cold
$\hat{R}$ = 2.882
Step

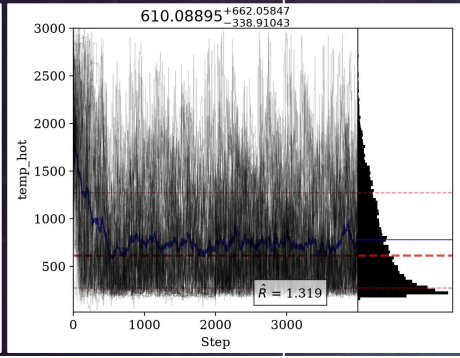

$-7.47725^{+0.13244}_{-0.11407}$
log_dust_mass_hot
$\hat{R}$ = 2.048
Step

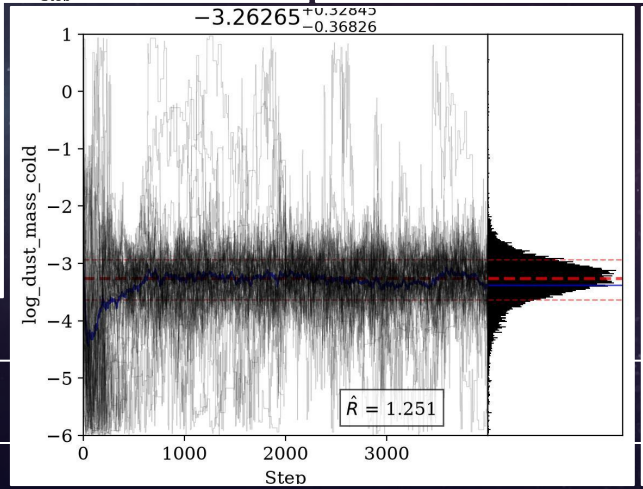
2C Carbon

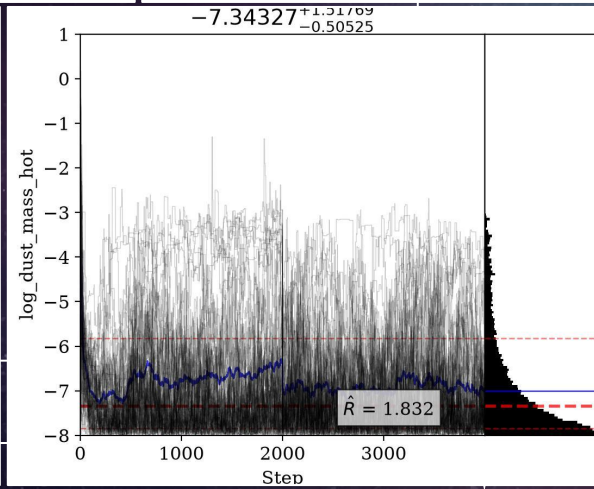

T_{cold} [K] = $188.824^{+20.772}_{-18.509}$
$\log(M_{hot}/M_\odot) = -7.343^{+1.502}_{-0.504}$
T_{hot} [K] = $610.089^{+658.618}_{-338.080}$
$\sigma = 1.148^{+0.143}_{-0.119}$
T_{cold} [K]
$\log(M_{hot}/M_\odot)$
T_{hot} [K]
σ


$1.14807^{+0.14537}_{-0.11904}$
sigma
$\hat{R}$ = 1.151
Step

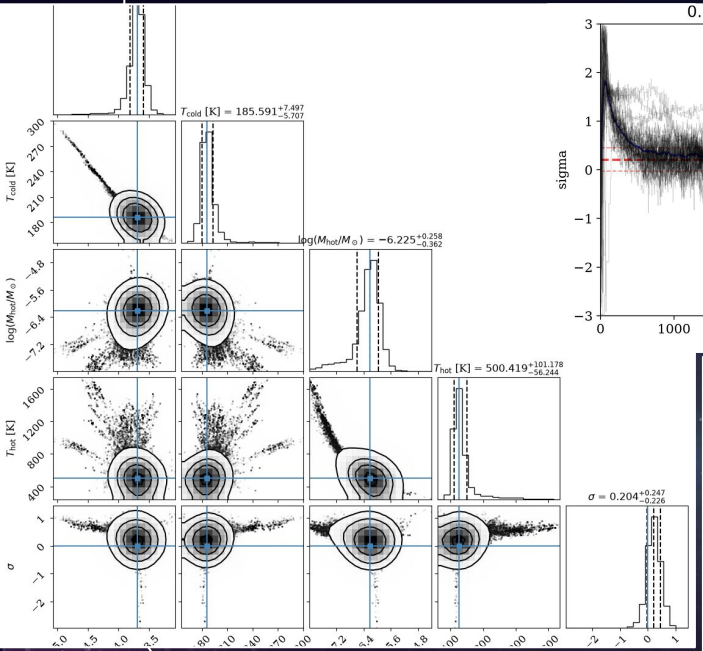

$188.82377^{+21.00864}_{-18.58079}$
temp_cold
$\hat{R}$ = 1.364
Step

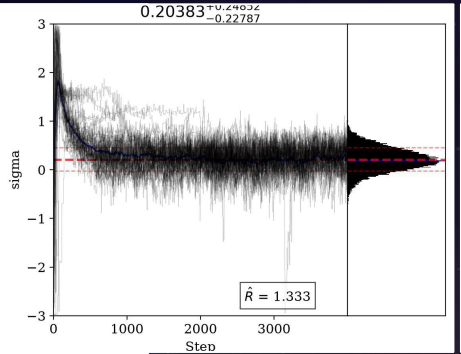

$610.08895^{+662.05847}_{-338.91043}$
temp_hot
$\hat{R}$ = 1.319
Step

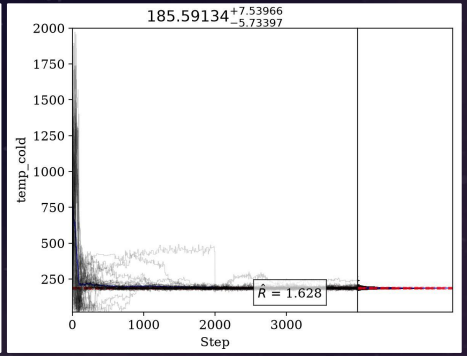

$-3.26265^{+0.32845}_{-0.36826}$
log_dust_mass_cold
$\hat{R}$ = 1.251
Step

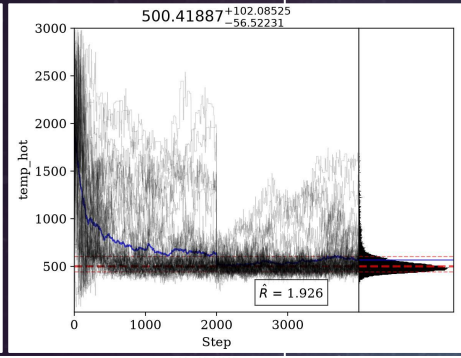

$-7.34327^{+1.51769}_{-0.50525}$
log_dust_mass_hot
$\hat{R}$ = 1.832
Step

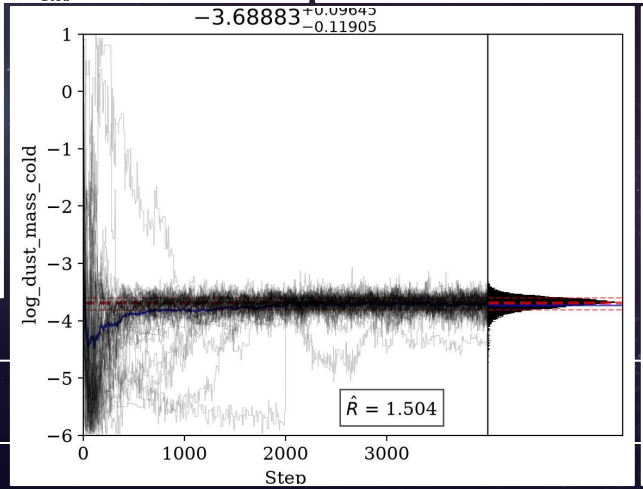
Cold Silicate, Hot Carbon

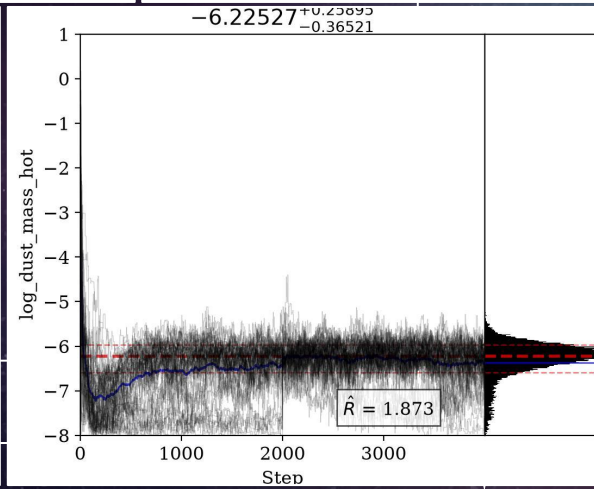




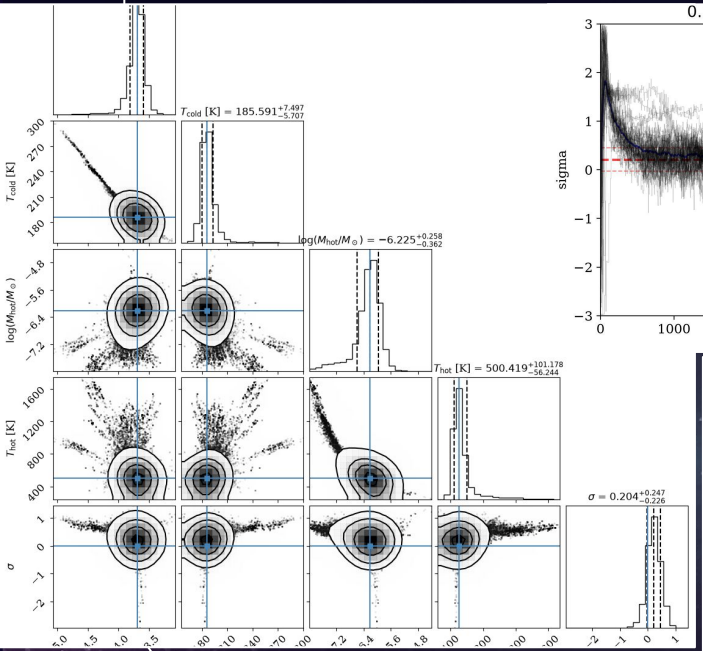


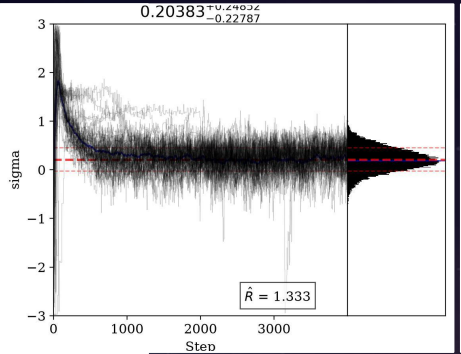


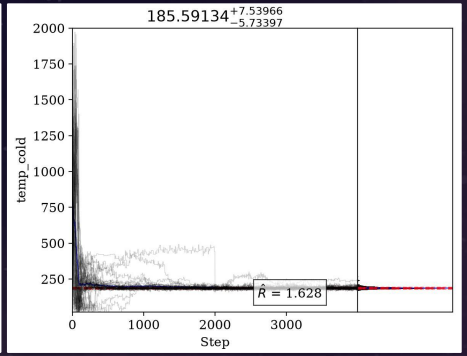


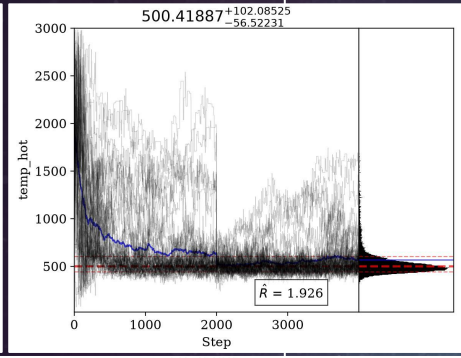


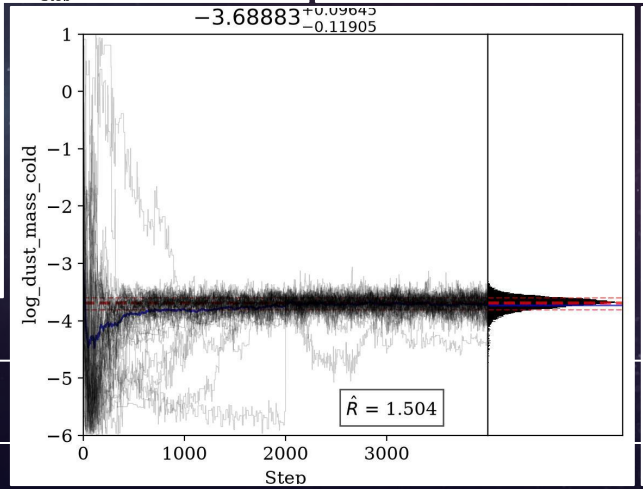
Cold Silicate, Warm Carbon

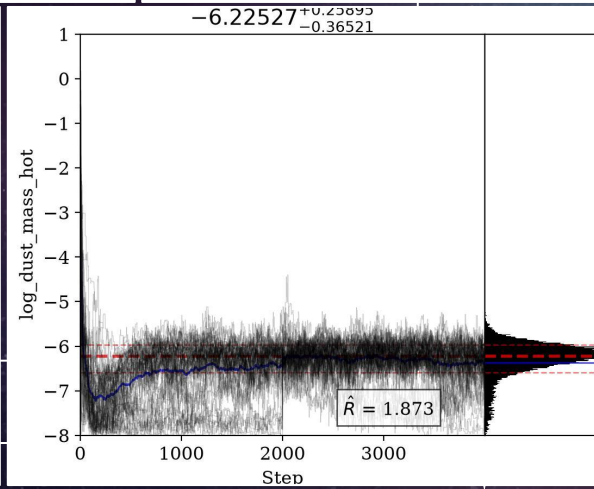




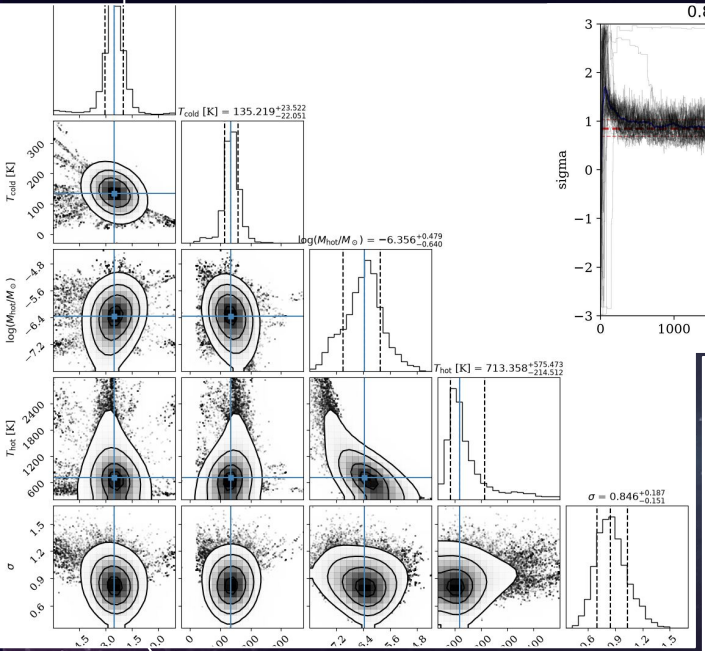


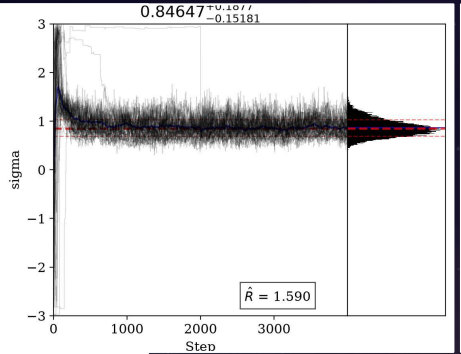


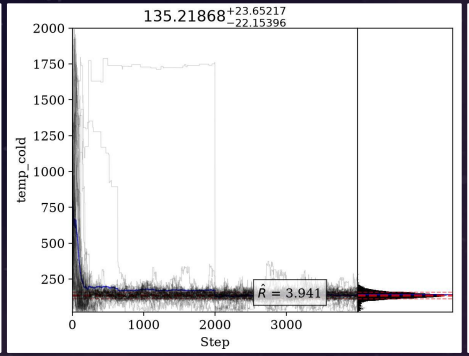


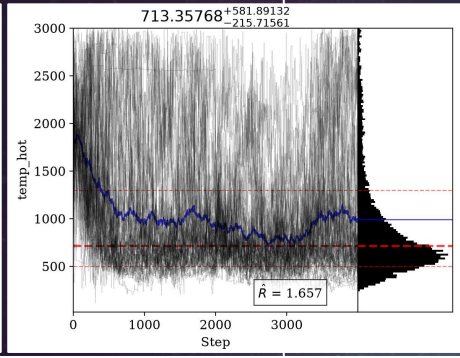


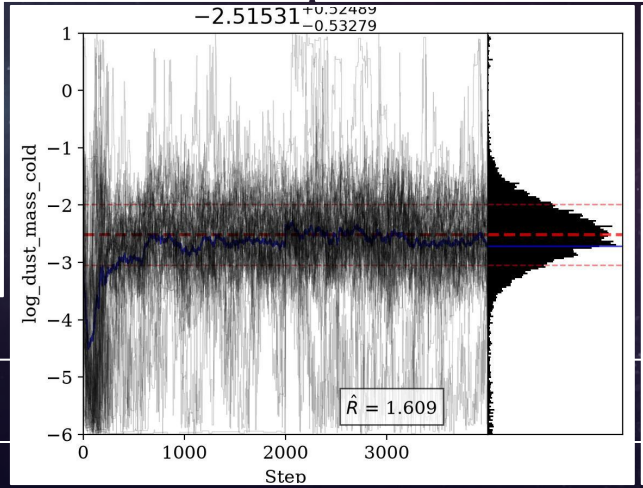
Hot Silicate, Cold Carbon

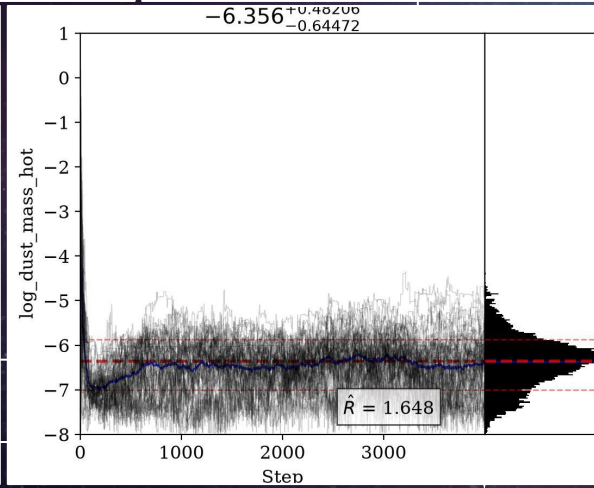












Dustysn 2017eaw fit outputs plugged into modified blackbody

